Atlantic wolffish (*Anarhichas lupus*)

Summary Scores

- **Sensitivity:** High
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	2.08	2.5	
Complexity in Reproductive Strategy	2.28	2.7	
Dispersal of Early Life Stages	2.24	2.8	
Egg and Larval Survival and Settlement Requirements	2.6	2.7	
Historical Variability in Temperature	3.04	2.9	
Juvenile Habitat Requirements	1.32	3	
Other Stressors	1.8	2.8	
Population Growth Rate	3.32	2.6	
Prey Specificity	2.36	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	2.48	2.7	
Spawning Cycle	2.72	2.8	
Stock Size/Status	3.52	2.4	
Total Sensitivity | Certainty	**High (3) | 0.99**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.34	1.13	
Bottom Oxygen	3.03	0.95	
Bottom Salinity	2.17	1.14	
Bottom Aragonite Solubility	1.16	0.48	
Net primary production	1.07	0.16	
Downward Flux of Particulate Organic Carbon	1.77	0.56	
Total Exposure	**Moderate (2.05)**	**0.92**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (6.16)**	**2.77**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Atlantic wolffish. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	High	High	None
Exposure	High	Moderate	Decrease
Vulnerability	High	Moderate	Decrease
Directional Effect	Negative	Negative	None

Stocks

Atlantic wolffish is managed as a single stock or stock information was not considered for this iteration of the CVA. ## Score Descriptions:

Sensitivity:

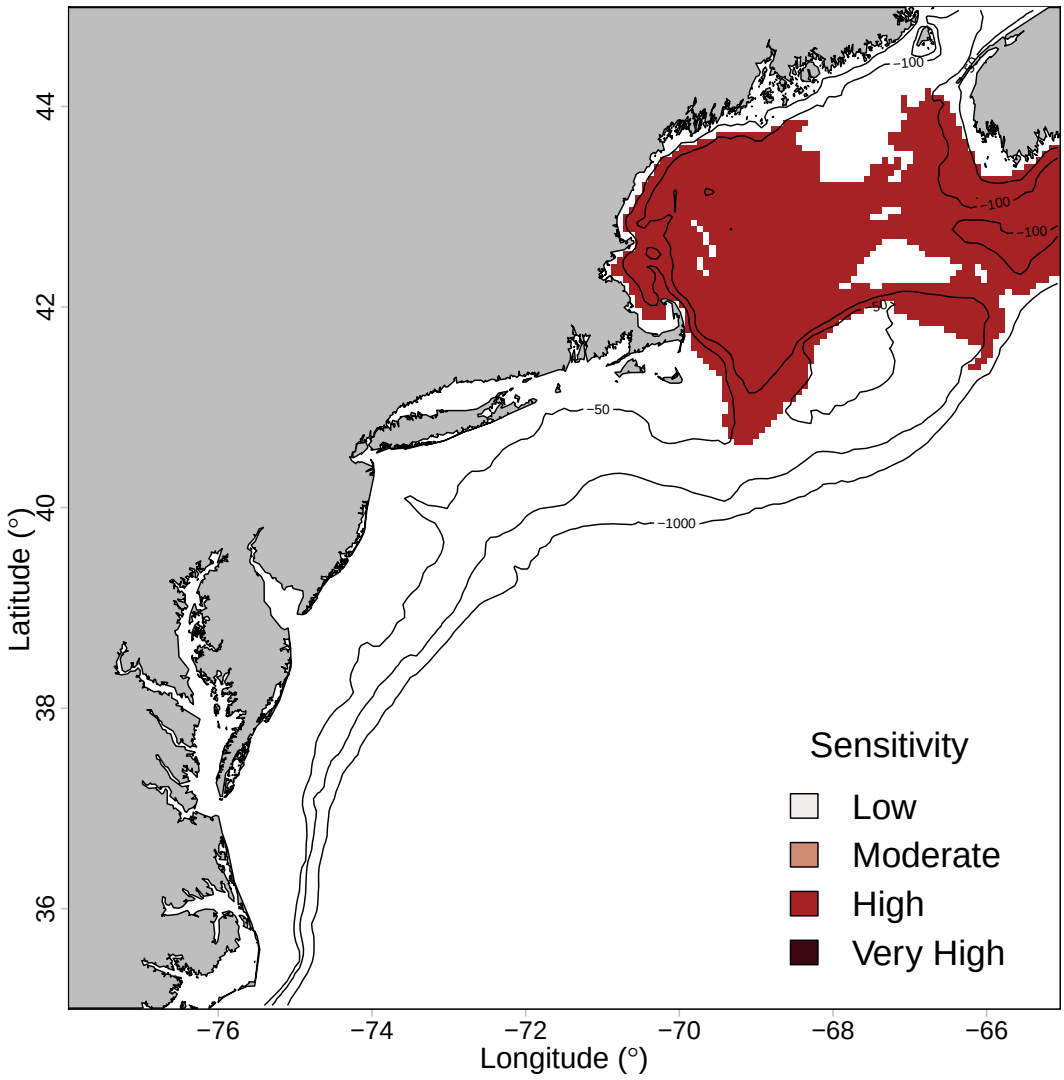


Figure 2: Atlantic wolffish Sensitivity on the Northeast Shelf

Atlantic Wolffish received a **High** sensitivity score. This score had a certainty of 99% from bootstrap analysis. Three attributes had scores greater than 3: Stock Size (3.5), Population Growth Rate (3.3), and Historical Variability in Temperature (3) (Figure 1).

Wolffish are a long lived (maximum documented age is 33 years) and late maturing species (most reach maturity by age 6) (Rountree 2002; McBride et al. 2022). Furthermore, wolffish prefer colder waters and previous studies have illustrated that they will shift to seek out preferred temperature (Kulka et al. 2011; Rountree 2002). Wolffish are not currently overfished and overfishing is not occurring (NEFSC 2022).

In the previous Climate Vulnerability Assessment, Atlantic wolffish also received a High sensitivity score (Hare et al. 2016). This was driven by Stock Status, Population Growth Rate, and Spawning Cycle, similar to our results. Previous research suggested that spawning only occurred in the autumn in the Northwest Atlantic, which likely drove the high Spawning Cycle score in the previous Assessment. However, more

recent work has shown that timing may vary according to local conditions (Rountree 2002; McBride et al. 2022).

Exposure:

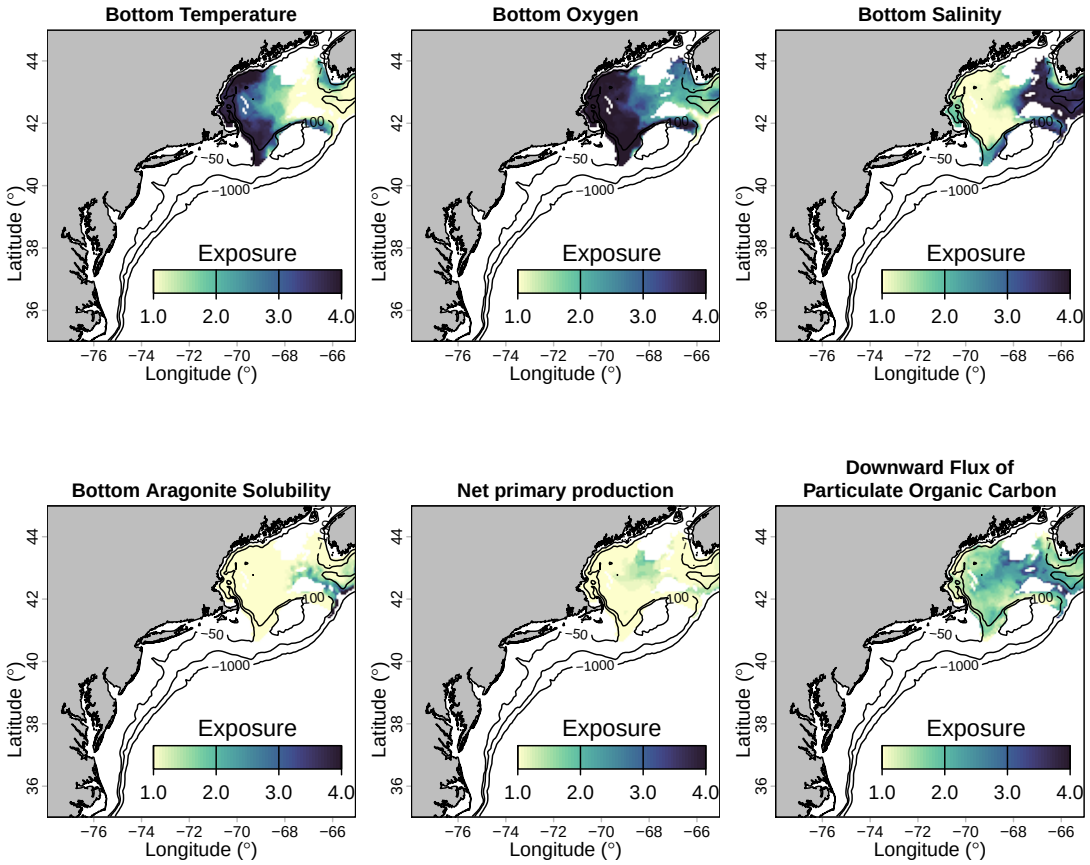


Figure 3: Maps of each Exposure Factor for Atlantic wolffish. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure for Atlantic Wolffish throughout its range was **Moderate** (2.05 +/- 0.92) (Figure 1). This was driven primarily by high exposure to changes in bottom temperature and oxygen (Figure 3). Exposure was highest along the western coast of the Gulf of Maine as well as on the northern flank of Georges Bank and east of Cape Cod (Figure 4).

In the previous Assessment, Atlantic Wolffish received a High exposure score, driven by sea surface temperature and ocean acidification (Hare et al. 2016).

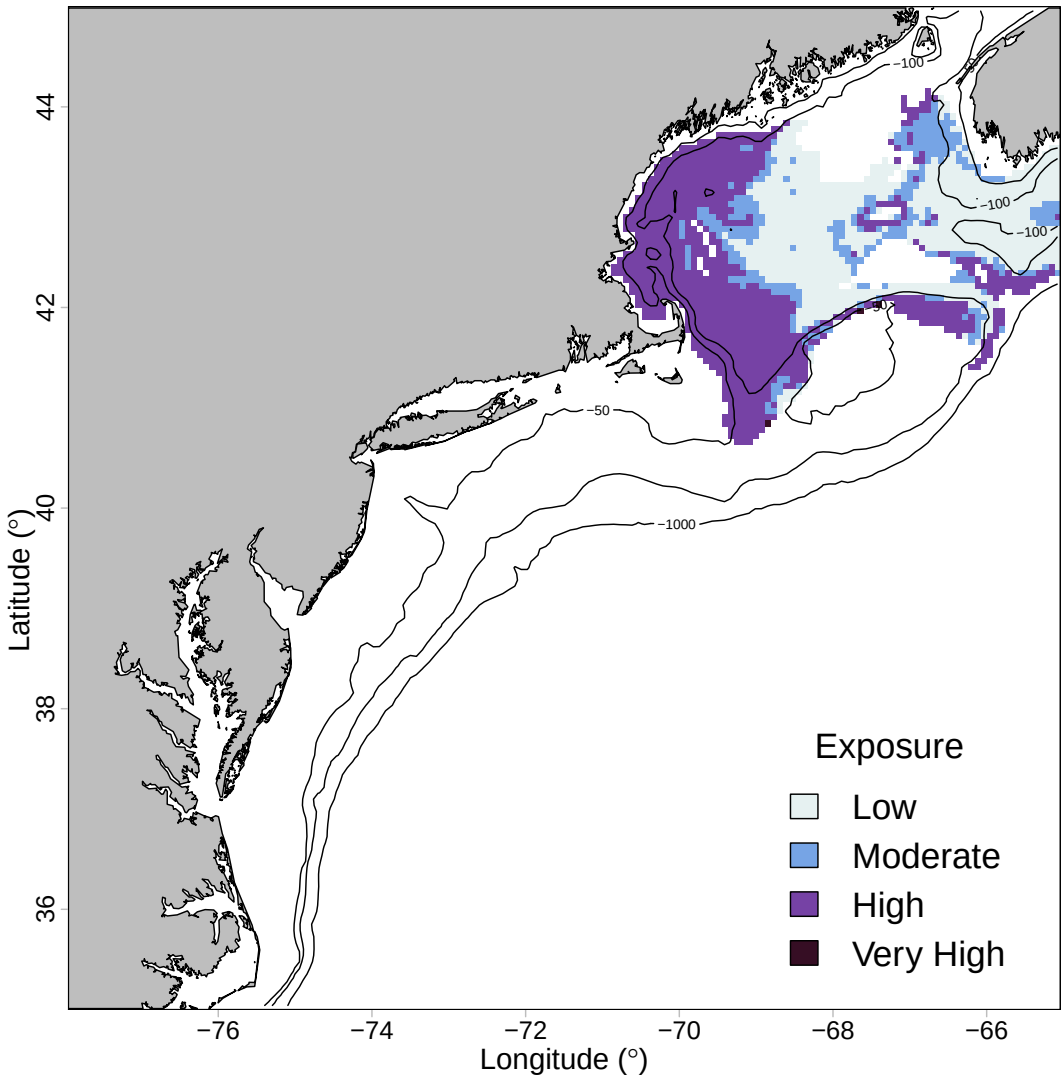


Figure 4: Total Exposure on the Northeast Shelf for Atlantic wolffish. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

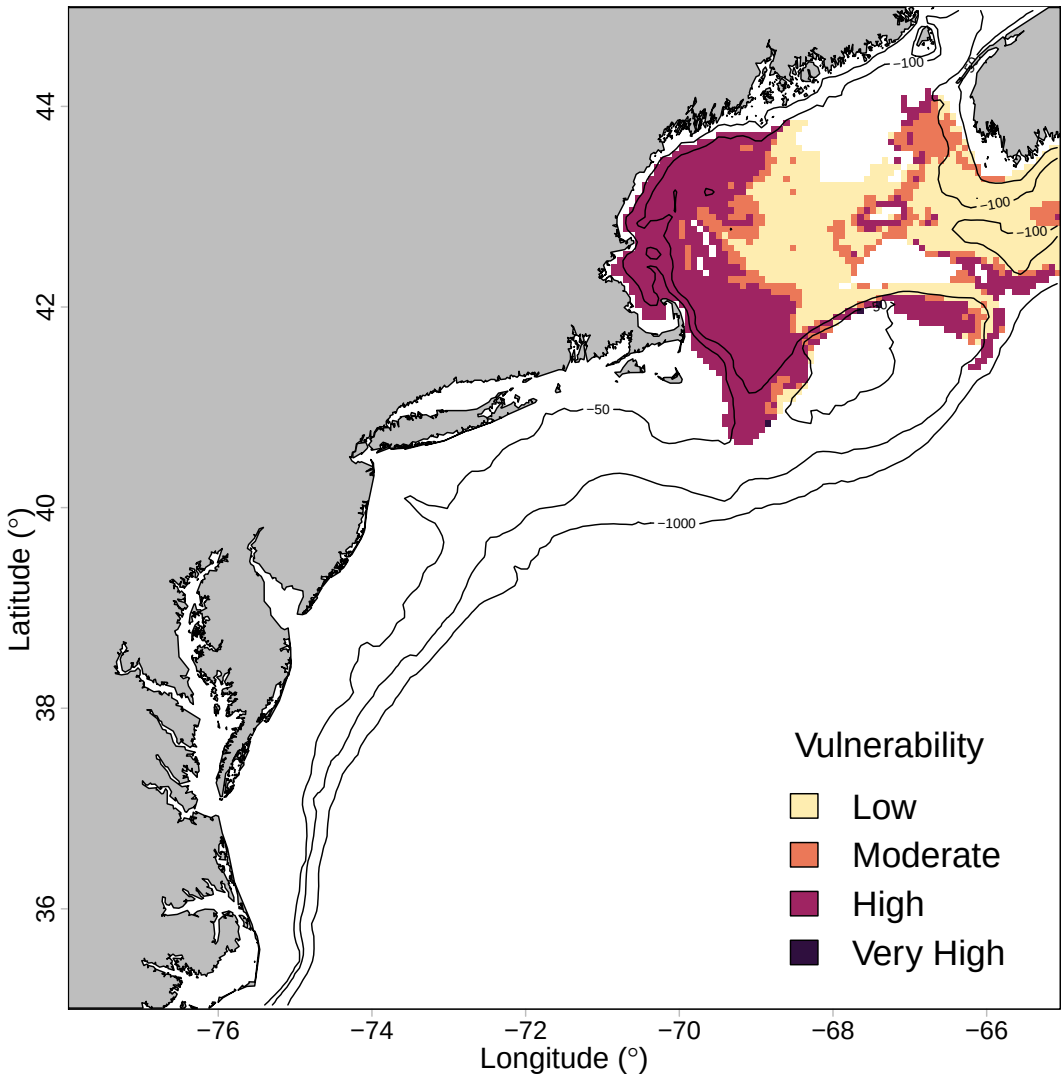


Figure 5: Vulnerability on the Northeast Shelf for Atlantic wolffish. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability for the Atlantic Wolffish throughout its range was **Moderate** (6.16 +/- 2.77) (Figure 1). Vulnerability was High along the western coast of the Gulf of Maine, and on the northern flank of Georges Bank and Low elsewhere throughout the Gulf of Maine (Figure 5).

In the previous Assessment, wolffish received a High Vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

97% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

The wolffish ensemble model received a model confidence score of 2.4 (+/- 0.1), indicating moderate confidence in the final model. Experts stated that the modeled distributions met expectations, but low sample size due to infrequent catches in the surveys lowered scores. The ensemble model AUC was 0.87, indicating good model performance.

Additional Metrics

Distribution Shifts:

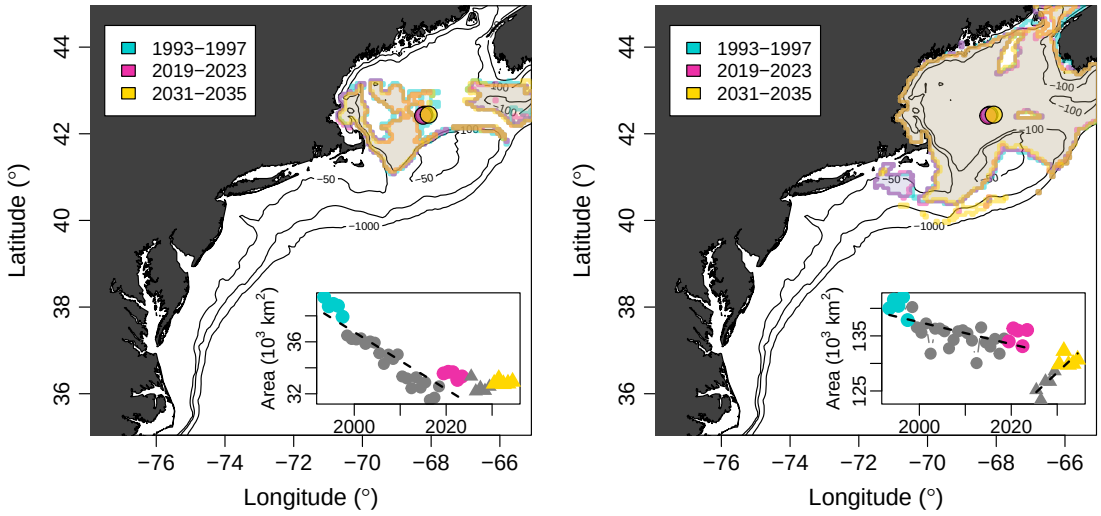


Figure 6: Distribution shifts on the Northeast Shelf for Atlantic wolffish. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

There were significant declines in habitat area for Atlantic wolffish due to a southward shift in the northern edge of the distribution (Figure 6). The weighted centroid shifted less than 5 km south-southwest between 1993 and 2023.

Distributions are predicted to remain similar to current conditions between 2025 and 2035. Wolffish are predicted to lose a portion of their habitat in southern New England, but gain more habitat near the shelf break (Figure 6). The weighted centroid is predicted to shift a further 14 km east.

In the previous Assessment, wolffish received a Moderate Distributional Vulnerability Rank due to their sedentary behaviors and preference for complex habitats (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	38,957	140,548	2.81	4.97
2019-2023	3.83 km SSW	33,572 (-5,385)	135,170 (-5,378)	2.62 (-0.19)	4.97 (0)
2031-2035	13.5 km E	32,880 (-692)	130,498 (-4,672)	2.49 (-0.12)	4.97 (0)

Directional Effect:

Atlantic wolffish are expected to be **negatively** impacted by future climate change. Certainty scores from experts were 100%. Atlantic wolffish are known temperature seekers and prefer cold waters (Kulka et al. 2011). Low dissolved oxygen concentrations may also impact wolffish (Bianucci et al. 2016). These results are similar to those from the 2016 Climate Vulnerability Assessment for the region (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Atlantic wolffish is a cold water species and has been termed a "temperature seeker" (Kulka et al. 2011). Early life history stages and fish near the southern stock boundary may be the most vulnerable to surface water temperatures exceeding 9-11°C (Rountree 2002). Juvenile Atlantic wolffish exposed to temperatures of 15°C exhibited reduced growth and higher metabolic rates compared to those at 10°C (Hinchcliffe et al. 2025). In the North Sea, Atlantic wolffish have shifted their distribution and decreased abundance at the southern end of their range and in shallow waters (Bluemel et al. 2022). Bianucci et al. (2016) observed a connection between low dissolved oxygen and contraction of the Atlantic wolffish population on the Scotian Shelf. Model results of future dissolved oxygen conditions also extend to areas with high Atlantic wolffish biomass (Bianucci et al. 2016). Similar results were also observed in (Brennan et al. 2016) which predicted declines in dissolved oxygen concentrations and increases in temperature outside of known adult wolffish tolerances.

Life History Synopsis

Atlantic wolffish inhabit deep, rocky waters of the Northwest Atlantic from Labrador to the Great South Channel and Georges Bank (Rountree 2002). In the Northwest Atlantic, wolffish are believed to spawn primarily in autumn (Rountree 2002). Spawning is likely temperature dependent (Atlantic Wolffish Biological Review Team 2009). Wolffish will migrate to colder temperatures before the spawning season, and egg laying and guarding behavior could be disrupted if bottom temperatures increase above 10°C (Collette and Klein-MacPhee 2002). However, laboratory experiments suggest that timing may be highly plastic and dependent on local conditions, including photoperiod (McBride et al. 2022). Spatial and temporal variation in spawning may be contributing to the stock's regional reproductive isolation (Atlantic Wolffish Biological Review Team 2009). Eggs are deposited in rocky dens and incubation time is highly dependent on water temperature, varying between 3 - 9 months (Atlantic Wolffish Biological Review Team 2009). After hatching, prolarvae stay close to the bottom. Larvae consume a broad range of amphipod, crustacean, and fish larvae (Rountree 2002). The prolarval stage can last from a few hours to 6 days

(Collette and Klein-MacPhee 2002). The larvae become pelagic shortly after and stay within the water column from 20 days to 2 months, with colder temperatures prolonging the larval period, before migrating back to the bottom (Rountree 2002). Juvenile and adult wolffish occupy the bottom habitats of the continental shelf and slope, using large stones and rocks for shelter and nesting and soft substrates for feeding (Atlantic Wolffish Biological Review Team 2009) (Collette and Klein-MacPhee 2002). Juvenile wolffish primarily consume echinoderms (Atlantic Wolffish Biological Review Team 2009). Adults prey upon hard-shelled invertebrates such as mollusks, crustaceans and echinoderms. Stomach contents have included sea urchins, whelks, cockles, sea clams, brittle stars, crabs, scallops and other shellfish in addition to an occasional redfish (Atlantic Wolffish Biological Review Team 2009) (Rountree 2002). While previous studies have suggested that wolffish migrate inshore/offshore seasonally for spawning or maturity, it is unclear if this is a true migration or if behavioral changes are affecting catchability as eggs are found both inshore and offshore. More recent tagging studies have illustrated that movements in the Northwest Atlantic are limited, but the species has been shown to migrate in the Northeast Atlantic (Rountree 2002) (Simpson et al. 2015) (Gunnarsson et al. 2019). Spiny dogfish, sea raven and Atlantic cod are the dominant predators for juveniles. For adults, the occurrence of predation is low and is not likely to be having a significant effect at the population level (Atlantic Wolffish Biological Review Team 2009). The maximum reported age for wolffish in the Gulf of Maine is 33 years (Fairchild et al. 2015). Most individuals reach maturity by age 6, but this may vary regionally due to temperature (Rountree 2002) (McBride et al. 2022). Atlantic wolffish are managed by the New England Fishery Management Council under the Northeast Multispecies (Groundfish) Fishery Management Plan. As of the 2022 Management Track Assessment, the stock is not overfished and overfishing is not occurring (Northeast Fisheries Science Center (NEFSC) 2022).

Supplemental Scores

Exposure Timeseries

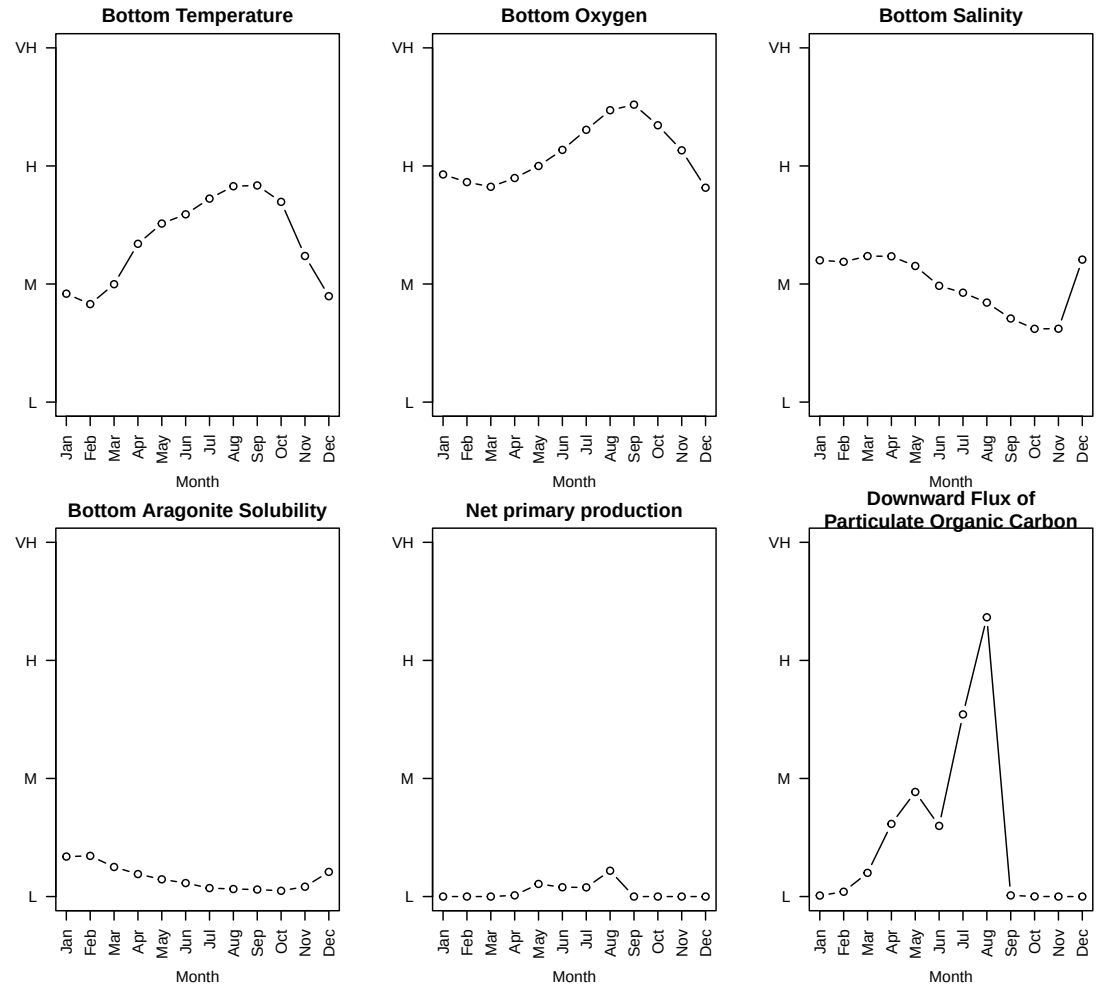


Figure 7: Monthly Timeseries of each Exposure Factor for Atlantic wolffish. Exposure was calculated between 2014-2023 and 2025-2035

Monthly exposure ranged from Low to High. Scores were Low from November through April, and increased to High in August (Figure 8). This was due to changes in bottom temperature and oxygen, as well as downward carbon flux exposure (Figure 7).

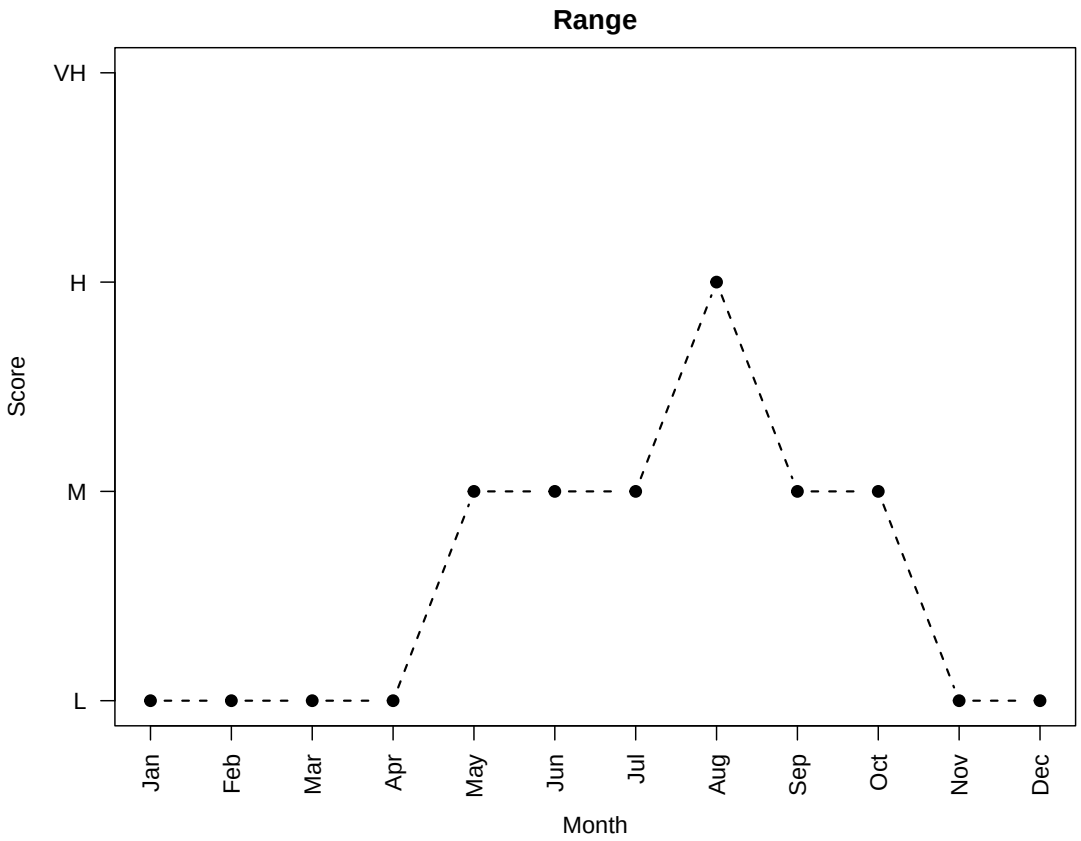


Figure 8: Timeseries of Total Exposure for Atlantic wolffish

Exposure due to Significant Factors

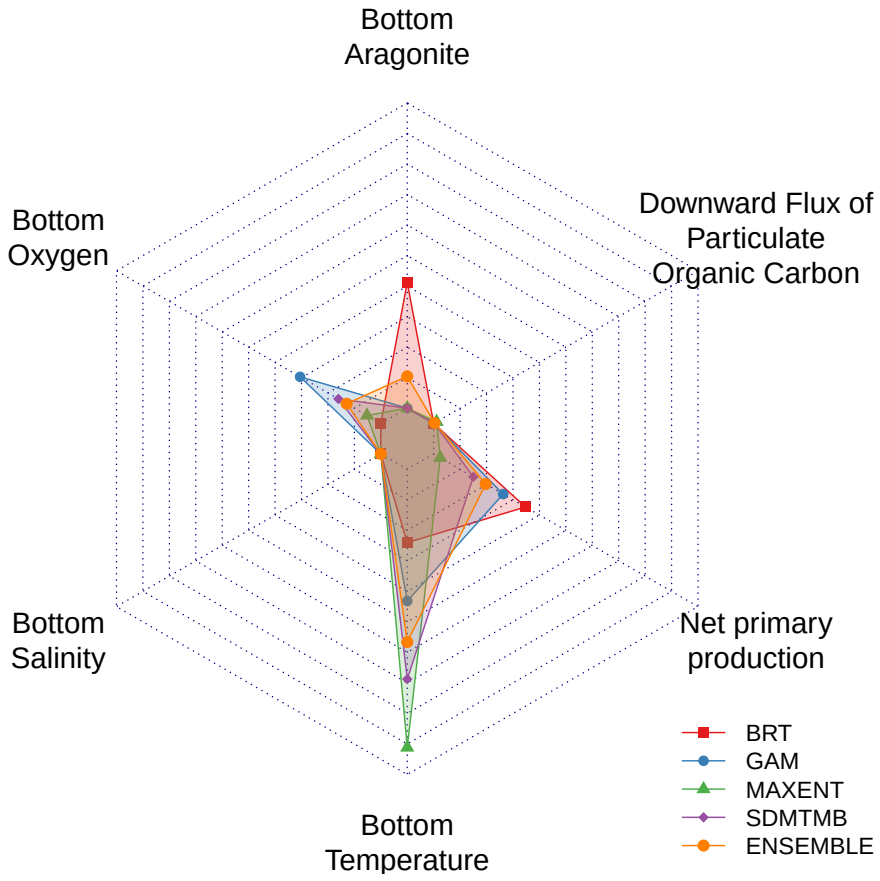


Figure 9: Dynamic Factors driving Atlantic wolffish distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, oxygen, aragonite solubility, as well as net primary production were the strongest dynamic drivers of wolffish distribution (Figure 9). Wolffish are described as 'temperature-seekers' and bottom oxygen concentrations are known drivers of species distributions (Kulka et al. 2011; Rountree 2002; Bianucci et al. 2016; Brennan et al. 2016). Both adults prey on a variety of bottom species, including mollusks, which may explain why bottom aragonite saturation state drove species distributions (Atlantic Wolffish Biological Review Team 2009; Rountree 2002). When Exposure was calculated using only these factors, Exposure remained Moderate (1.81 +/- 0.94) and spatial patterns were similar (Figure 11; Figure 10).

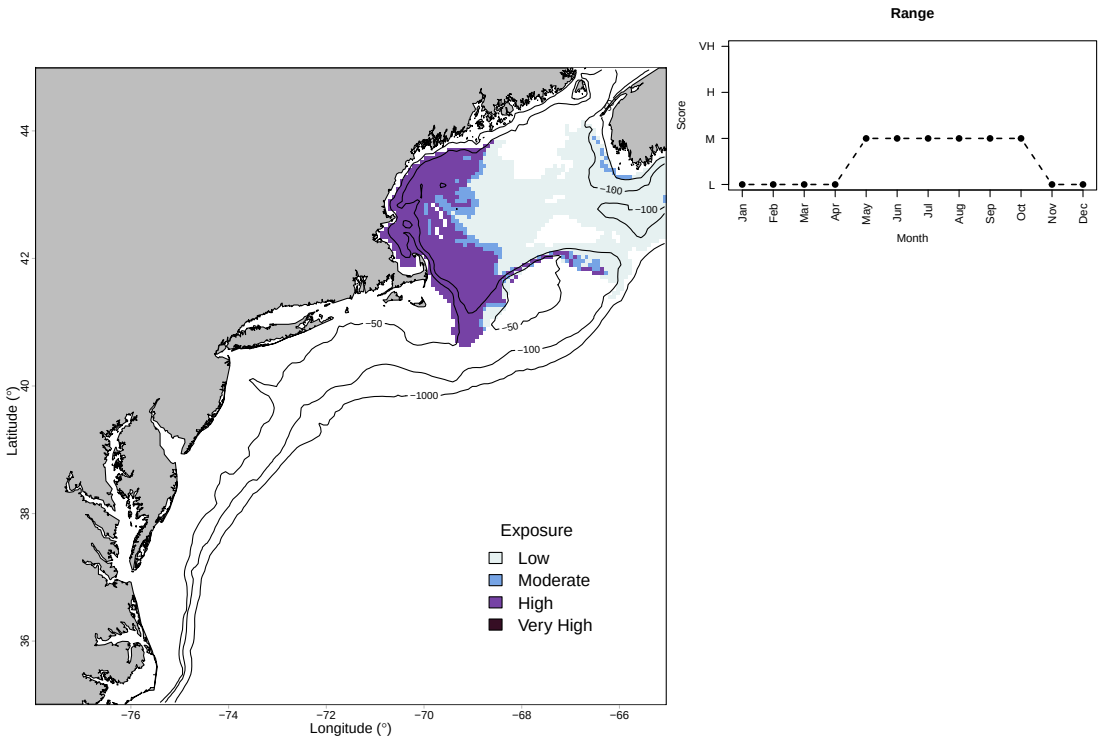


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Atlantic wolffish distributions

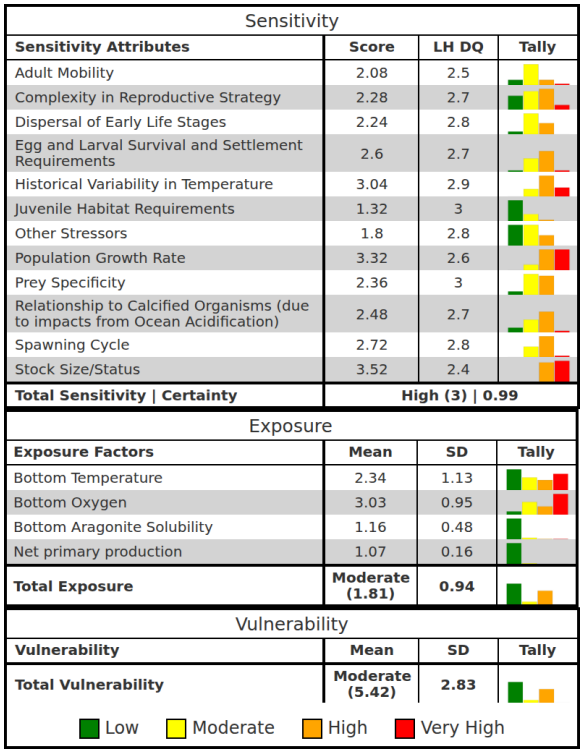


Figure 11: As in Figure 1 for Atlantic wolffish, but with only significant Exposure factors and the resulting Vulnerability Scores.

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